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Assignment - 2

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Subject - Statistics and Probability (MAS 2001)

Section - I

Q1 (i) If $n=5$, $p=.20$ then what will be the mean and varience of Binomial distribution.

$$\text{mean} = np = 5 \times 0.20 = \boxed{1}$$

$$\text{Varience} = npq = np(1-p) = 5 \times 0.20 \times 0.80 = \boxed{0.8}$$

(ii) The mean and varience of Binomial distribution are 4 and 4/3 respectively. Find $P(X \geq 1)$

$$\mu = np = 4 \quad -①$$

$$\sigma^2 = npq = \frac{4}{3}$$

$$4q = \frac{4}{3}$$

$$\boxed{q = \frac{1}{3}}, \quad \boxed{p = 1-q = \frac{2}{3}}$$

$$; \quad np = 4$$

$$n\frac{2}{3} = 4$$

$$\boxed{n = 6}$$

$$P(X \geq 1) = 1 - P(X<1)$$

$$= 1 - P(X=0)$$

$$= 1 - {}^nC_x\, p^x q^{n-x}$$

$$= 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{6-0}$$

$$= 1 - \left(\frac{1}{3}\right)^6$$

$$= \frac{728}{729}$$

$$\boxed{P(X \geq 1) = 0.99863} \text{ Ans}$$

(iii) Comment on the following statement: "The mean of a Binomial distribution is 5 and standard deviation is 3"

$$\mu = 5 \quad ; \quad \sigma = 3$$

$$np = 5 \quad ; \quad \sigma^2 = 9$$

$$npq = 9$$

$5q = 9$

$q = \frac{9}{5} > 1$

$\because q \le 1$, ie the statement is incorrect.

(10) If X is a Poisson variate such that

$$P(X=2) = 9P(X=4) + 90P(X=6)$$

find the mean and variance of X.

Sol Poission law ⇒

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad ; x \ge 0 \quad , \text{where } \lambda > 0$$

$$P(X=2) = 9P(X=4) + 90P(X=6)$$

$$\frac{\cancel{e^{-\lambda}}\lambda^2}{2!} = \frac{9\cancel{e^{-\lambda}}\lambda^4}{4!} + \frac{90\cancel{e^{-\lambda}}\lambda^6}{6!}$$

$$\frac{1}{2} = \frac{3}{8}\lambda^2 + \frac{\lambda^4}{8}$$

$$4 = 3\lambda^2 + \lambda^4$$

let $\lambda^2 = x$

$\Rightarrow x^2 + 3x - 4 = 0$

$\Rightarrow (x+4)(x-1) = 0$

$\Rightarrow \boxed{x = -4}, \boxed{x = 1}$

$\lambda^2 = -4$

$\lambda = \sqrt{-4}$

(can't be imaginary) ✗

$\lambda^2 = 1$

$\lambda = \pm 1$

$\because \lambda > 0$

ie $\boxed{\lambda = 1}$

As in poisson distribution law ⇒

$$\boxed{\begin{aligned} \text{mean} &= \lambda = 1 \\ \text{variance} &= \lambda = 1 \end{aligned}}$$

Ans

(vi) Under normal curve total area right side of mean is $\underline{50\%}$

(vii) Mean and variance of standard normal variate are ____

$\mu = 0, \quad \sigma^2 = 1$

(viii) What is the p.d.f. the Uniform distribution,

$$\text{p.d.f} = f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{elsewhere} \end{cases}$$

(ix) The mean and variance of Uniform distribution are 1 and 4/3 respectively. Find $P(X<0)$

$$\text{mean} = \mu = \frac{a+b}{2} = 1$$

$$\boxed{a+b = 2}$$

$$\text{variance} = \sigma^2 = \frac{(b-a)^2}{12} = \frac{4}{3}$$

$$(b-a) = \pm 4$$

$$b - a = 4 \qquad\qquad b - a = -4$$

$$+\ a + b = 2 \qquad\qquad +\ a + b = 2$$

$$2b = 6 \qquad\qquad 2b = -2$$

$$\boxed{b = 3} \qquad\qquad \boxed{b = 1}$$

$$\boxed{a = -1} \checkmark \qquad\qquad \boxed{a = 3}$$

✗ Not possible as $a < b$ is must condition

$$f(x) = \begin{cases} \frac{1}{b-a} = \frac{1}{3-(-1)} = \frac{1}{4}; & (-1 < x < 3) \\ 0; & \text{elsewhere} \end{cases}$$

$$P(X<0) = \int_{-\infty}^{0} f(x)\,dx = \int_{-1}^{0} f(x)\,dx = \int_{-1}^{0} \frac{1}{4}\,dx = \frac{1}{4}[0-(-1)]$$

$$\boxed{P(X<0) = \frac{1}{4}} \quad \text{Ans}$$

(x) What is the p.d.f of exponential distribution.

$$P.DF = f(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{elsewhere} \end{cases}$$

(xi) In which condition mean and variance of exponential distribution are equal.

$$\text{mean} = \frac{1}{\lambda}$$

$$\text{variance} = \frac{1}{\lambda^2}$$

Possible if $\boxed{\lambda = 1}$ Ans

(xii) Which continuous distribution follows memoryless property

↳ exponential distribution.

Q.2 Out of 800 familie with 4 children each, how many families would be expected to have

(i) 2 boys and 2 girls

(ii) at least one boy

(iii) no girl

(iv) at most two girls.?

Sol

$n = 4,$

$p = \frac{1}{2}$

$q = \frac{1}{2}$

(i) $P(X=2) = {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2$

$= \frac{4 \times 3}{2} \times \frac{1}{4} \times \frac{1}{4} = 0.375$

$P(X=2)$ for 800 families $= 800 \times 0.375 = \boxed{300}$ Ans

(ii) $P(X \geq 1) = 1 - P(X=0)$ ← no Boy

$= 1 - {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4$

$= 1 - \frac{1}{16}$

$= 0.9375$

for 800 families

$P(X \geq 1) = 800 \times 0.9375 = \boxed{750}$ Ans

(iii) no girl ⇒ or all Boy

$P(X=4) = P(X=0) = {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4$

$= \left(\frac{1}{2}\right)^4$

$= 0.0625$

for 800 families

$P(X=4) = 800 \times 0.0625 = \boxed{50}$ Ans

(iv) atleast two girls ⇒

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2$

$= \frac{1}{16} + \frac{1}{4} + \frac{3}{8}$

$= 0.6875$

for 800 families

$P(X \leq 2) = 800 \times 0.6875 = \boxed{550}$ Ans

Q.3 Twenty percent screw produced in a certain factory turn out to be defective. Find the probability that in a sample of 15 screws chosen at random exactly four will be defective.

Ans

$p = 0.8$ (not defective)

$q = 0.2$ (defective)

$n = 15$

Four defective, ie 11 not defective $\Rightarrow$

So $P(X=11) = {}^{15}C_{11} (0.8)^{11} (0.2)^{15-11}$

$= {}^{15}C_4 (0.8)^{11} (0.2)^4$

$= \boxed{0.1876}$ Ans

$\boxed{18.\%} =$

Q4 The probability that a bomb dropped from a plane will strike the target is 1/5. If six bombs are dropped find the probability that at least two will strike the target and exactly two will strike the target

Sol ⇒ (I) $p = \frac{1}{5}$, (using binomial distribution)

$q = 1 - \frac{1}{5} = \frac{4}{5}$

$n = 6$

(1) atleast two will strike ⇒ $P(x \geq 2)$

$P(X \geq 2) = 1 - P(X<2)$

$= 1 - P(X=0) - P(X=1)$

$= 1 - {}^6C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^6 - {}^6C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^5$

$= 1 - \left(\frac{4}{5}\right)^6 - 6 \times \frac{1}{5} \times \left(\frac{4}{5}\right)^6$

$= \boxed{0.34464}$ Ans

(ii) exactly two will strike

$P(X=2) = {}^6C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^6$

$= \frac{6 \times 5}{2 \times 1} \times \frac{1}{5^2} + \frac{4^6}{5^6}$

$= \boxed{0.24576}$ A

$\boxed{Q5}$ In a book of ~~100 cellphones~~, 520 pages, 390 type-graphical errors occur. Assuming Poisson law for the number of errors per page., find the probability that a random sample of 5 pages will contain no error.

Sol

λ = avg success = avg num of error per page

$\lambda = \frac{390}{520} = 0.75$

λ for sample of 5 page $= 5 \times 0.75$

$\boxed{\lambda_{sample} = 3.75}$

→ Poisson law → $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$ $(x \geq 0)$

So no Error → $P(X=0) = \frac{e^{-3.75} \lambda^0}{0!} = \boxed{e^{-3.75}}$ Ans

Q-6 Six coins are tossed 6400 times. Using the Poisson distribution, find the approximate probability of getting 6 heads 2 times.

Sol

Poisson distribution →

$$PMF = P(X=x) = \left\{ \frac{e^{-\lambda}\lambda^{x}}{x!} \right\} \quad ; \text{ for } x \geq 0$$

* probability of getting 1 head $= \frac{1}{2}$
* probability of getting 6 head in 6 toss $= \left(\frac{1}{2}\right)^6$
* λ_{sample} = avg num of success $= 6400 \times \left(\frac{1}{2}\right)^6$
* $x = 2$

So $P(X=2) = \dfrac{e^{-\left(\frac{6400}{2^6}\right)} \left(\frac{6400}{2^6}\right)^2}{2!}$

$= \dfrac{e^{-100}(10000)}{2}$

$\boxed{P(X=2) = e^{-100} \cdot 5000}$ Ans

Q-7 A box contain 100 cellphones, 20 of which are defective. Then ten cellphones have been selected for inspection. Find the probability that

(i) at least one is defective
(ii) at the most three are defective
(iii) all are defective
(iv) none is defective

Sol

Total = 100

Defective = 20

Not defective = 80

Select 10 →

Sol ① at least one defective $= P(X \geq 1)$

$= 1 - P(X < 1)$

$= 1 - \underbrace{P(X=0)}_{\text{zero defective.}}$

$= 1 - \dfrac{{}^{80}C_{10}\ {}^{20}C_0}{{}^{100}C_{10}}$

$= 1 - 0.0951$

$= \boxed{0.9048}$ Ans

(ii) at most three defective ⇒

$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= \dfrac{{}^{80}C_{10}}{{}^{100}C_{10}} + \dfrac{{}^{80}C_9\ {}^{20}C_1}{{}^{100}C_{10}} + \dfrac{{}^{80}C_8\ {}^{20}C_2}{{}^{100}C_{10}} + \dfrac{{}^{80}C_7\ {}^{20}C_3}{{}^{100}C_{10}}$

$= \boxed{0.8904}$ Ans

(iii) all are defective ⇒

$P(X=10) = \dfrac{{}^{80}C_0\ {}^{20}C_{10}}{{}^{100}C_{10}} = \boxed{1.0673 \times 10^{-8}}$ Ans

(iv) none are defective ⇒

$P(X=0) = \dfrac{{}^{80}C_{10}\ {}^{20}C_0}{{}^{100}C_{10}} = \boxed{0.0951}$ Ans

Q-8 In a certain examination, 2000 students appeared. The average marks obtained were 50% and standard deviation was 5%. How many student do you expect to obtain more than 60 marks, assuming the marks to be distributed normaly

Sol given $\mu = 50$

$\sigma = 5$

to find $P(x \geq 60) = ?$

$\Rightarrow P(x \geq 60)$

$\Rightarrow P\left(\frac{x-\mu}{\sigma} \geq \frac{60-\mu}{\sigma}\right)$

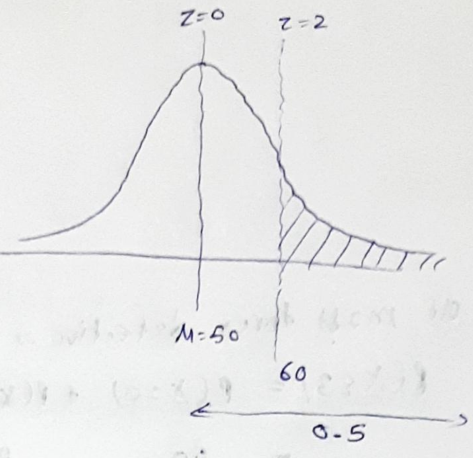
$\Rightarrow P\left(z \geq \frac{60-50}{5}\right)$

$\Rightarrow P(z \geq 2)$

$\Rightarrow 0.5 - P(0 < z < 2)$

$\Rightarrow 0.5 - 0.4772$

$P(x > 60) \Rightarrow \underline{0.0228}$



for 2000 Students $\Rightarrow$

$P(x > 60) = 2000 \times 0.0228$

$= 45.6$

$= \boxed{46 \text{ marks}}$ Ans

Q-9 Subway trains on a certain line run every half hour between midnight and six in the morning. What is the probability that a man entering the station at random time during this period will have to wait at least twenty minutes

Sol let $x \to$ uniformly distributed

$x \in (0, 30)$

$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{elsewhere} \end{cases}$

$\Rightarrow f(x) = \begin{cases} \frac{1}{30}, & 0 < x < 30 \\ 0, & \text{elsewhere} \end{cases}$

* wait at least 20 minutes $\Rightarrow$

$P(x \geq 20) = 1 - P(x < 20)$

$= 1 - P(0 < x < 20)$

$$= 1 - \int_{a}^{b} f(x)\,dx$$

$$= 1 - \int_{0}^{20} \frac{1}{30}\,dx$$

$$= 1 - \frac{1}{30}[20-0]$$

$$= 1 - \frac{2}{3}$$

$$= \frac{1}{3}$$

So man has probability $= \frac{1}{3}$ to wait atleast 20 minutes

Q-10 A random sample of size 100 is taken from an infinite population having mean 76 and varience 256. What is probability that sample mean will between 75 and 78?

Sol given, $n = 100$

$\mu = 76$

$\sigma^2 = 256$

$\Rightarrow \sigma = 16$

to find $P(75 < \bar{x}\ 78) = ?$

using CLT $\Rightarrow$

$$P(75 \le \bar{x} < 78) \Rightarrow P(75-\mu < \bar{x}-\mu < 78-\mu)$$

$$= P\left(\frac{75-\mu}{\frac{\sigma}{\sqrt{n}}} < \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} < \frac{78-\mu}{\frac{\sigma}{\sqrt{n}}}\right)$$

$$= P\left(\frac{75-76}{\frac{16}{10}} < Z < \frac{78-76}{\frac{16}{10}}\right)$$

$$= P(-0.625 < Z < 1.25)$$

$$= P(0 < Z < 0.625) + P(0 < Z < 1.25)$$

$$= P(0 < Z < 0.63) + P(0 < Z < 1.25)$$

$$= 0.2357 + 0.3944$$

$$P(75 < \bar{x} < 78) = \boxed{0.6301}$$ Ans

Q11 A car hire firm has two cars, which it hires out day by day. The number of demands for a car each day distributed as a Poisson distribution with mean 1.5, calculate the proportion of days on which (i) neither car is used and (ii) the proportion of days on which some demand refuses

Sol given mean $= 1.5 = \lambda$

$\Rightarrow \boxed{\lambda = 1.5}$

Poission distribution PMF $\Rightarrow$

$$PMF = P(X=x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!} & ; x = 0,1,2,\dots \\ 0 & ; \text{elsewhere} \end{cases}$$

(i) neither car is used $\Rightarrow$

$$P(X=0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} = \boxed{0.2231} \text{ Ans}$$

(ii) $P(X>2) = 1 - P(X \le 2)$

$= 1 - P(X=0) - P(X=1) - P(X=2)$

$= 1 - \frac{e^{-1.5}(1.5)^0}{0!} - \frac{e^{-1.5}(1.5)^1}{1!} - \frac{e^{-1.5}(1.5)^2}{2!}$

$= \boxed{0.19115}$ Ans

Q12 Assume mean height of soldiers to be 68.22 inches with a variance of 10.8 inches square. How many soldiers in a regiment of 1000 would you expect to be over 6 feet tall, given that the area under the standard normal curve between $x=0$ and $x=0.35$ is 0.1368 and between $x=0$ and $x=1.15$ is 0.3746.

Soln given $\mu = 68.22$ (6feet = 72 Inch)

$\sigma^2 = 10.8$

$\Rightarrow \sigma = \sqrt{10.8}$

to find ⇒

$$P(X > 72) = P\left(\frac{X-\mu}{\sigma} > \frac{72-\mu}{\sigma}\right)$$

$$\Rightarrow \quad = P\left(Z > \frac{72-68.22}{\sqrt{10.8}}\right)$$

$$= P(Z > 1.15)$$

$$= 0.5 - P(0 < Z\ 1.15)$$

$$= 0.5 - 0.3746$$

$$= 0.1254$$

for 1000 soldiers ⇒ $P(X > 72) = 1000 \times 0.1254$

$$\simeq \boxed{125 \text{ Soldiers}} \quad \text{Ans}$$

(Q-13) If X is Uniformly distributed in $[-2, 2]$

find (i) $P(X<0)$

(ii) $P(|X-1| \geq 1/2)$

Sol

~~(i) P(X<0) =~~

$x \in [-2, 2]$

$$\text{PDF} = f(x) = \begin{cases} \frac{1}{2-(-2)} = \frac{1}{4} ; & -2 < x < 2 \\ 0 ; & \text{elsewhere} \end{cases}$$

(i) $P(X<0) = \int_{-2}^{0} \left(\frac{1}{4}\right) dx$

$$= \frac{1}{4}(0-(-2))$$

$$= \boxed{\frac{1}{2}} \quad \text{Ans}$$

(ii) $P(|X-1| \geq \frac{1}{2}) = 1 - P(|X-1| < \frac{1}{2})$

$$= 1 - P\left(-\frac{1}{2} < X-1 < \frac{1}{2}\right)$$

$$= 1 - P\left(\frac{1}{2} < x < \frac{3}{2}\right)$$

$$= 1 - \int_{1/2}^{3/2} \left(\frac{1}{4}\right) dx$$

$$= 1 - \frac{1}{4}\left[\frac{3}{2} - \frac{1}{2}\right] = 1 - \frac{1}{4} = \boxed{\frac{3}{4}} \quad \text{Ans}$$

Q-14 The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = 1/3$. What is the probability that the repair time exceeds 3 hours?

Sol

given $\lambda = \frac{1}{3}$

exponential distribution → PDF ⇒

$$f(x) = \begin{cases} e^{-\lambda x}\lambda, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

to find $P(X > 3) = ?$

$$P(X>3) = 1 - P(X \leq 3)$$

$$= 1 - \int_0^3 e^{-(1/3)x} \left(\frac{1}{3}\right) dx$$

$$= 1 - \frac{1}{3}\left[\frac{e^{-x/3}}{-1/3}\right]_0^3$$

$$= 1 + \left[e^{-3/3} - e^{0}\right]$$

$$= 1 + e^{-1} - 1$$

$$= e^{-1}$$

$$\boxed{P(X>3) = 0.36787}$$ Ans

Q-15 The life length (in months) X of an electric component follow an exponentially distribution with parameter $\lambda = \frac{1}{2}$. What is the probability that the component survives at least 10 months given that already it had ~~the~~ survived for more than 9 months.

Sol

given exponential distribution ⇒

& PDF $= f(x) = \begin{cases} e^{-\lambda x}\lambda, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$

& $\lambda = \frac{1}{2}$

To find $\Rightarrow$ $P(x \geq 10 \mid \underbrace{x > 9}_{\text{already occured}}) = \dfrac{P(x \geq 10 \cap x > 9)}{P(x > 9)}$

$$= \frac{P(x \geq 10)}{P(x > 9)}$$

$$= \frac{1 - P(x < 10)}{1 - P(x < 9)}$$

$$= \frac{1 - \int_0^{10} e^{-(\frac{1}{2})x} \left(\frac{1}{2}\right) dx}{1 - \int_0^{9} e^{-(\frac{1}{2})x} \left(\frac{1}{2}\right) dx}$$

$$= \frac{1 - \frac{1}{2}\left[\frac{e^{-x/2}}{-1/2}\right]_0^{10}}{1 - \frac{1}{2}\left[\frac{e^{-x/2}}{-1/2}\right]_0^{9}}$$

$$= \frac{1 + (e^{-10/2} - e^{-0/2})}{1 + (e^{-9/2} - e^{-0/2})}$$

$$= \frac{1 + e^{-5} - 1}{1 + e^{-4.5} - 1}$$

$$= e^{-5} / e^{-4.5}$$

$$= \boxed{0.6065} \quad \text{Ans}$$

10-17 In a normal distribution, 31% of items are under 45 and 8% are over 64. What is the mean of standard deviation of the distribution?

Given that $\phi(1.41) = 0.42$

~~$\phi(0.19) = 0.50$~~

$\phi(0.5) = 0.19$

given,

$$P(X>64) = P\left(\frac{X-\mu}{\sigma} > \frac{64-\mu}{\sigma}\right) = 0.08$$

$$= P\left(z > \frac{64-\mu}{\sigma}\right) = 0.08$$

$$= 0.5 - P\left(0<z<\frac{64-\mu}{\sigma}\right) = 0.08$$

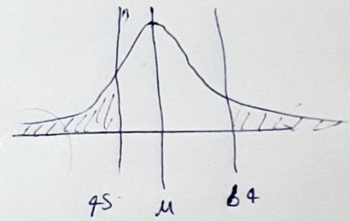
$$\Rightarrow P\left(0<z<\frac{64-\mu}{\sigma}\right) = 0.5-0.08$$

$$P\left(0<z<\frac{64-\mu}{\sigma}\right) = 0.42$$

in table area $=0.42$ corresponds to z score 1.41

ie $$\boxed{\frac{64-\mu}{\sigma} = 1.41} \text{ --- } (1)$$

given $P(X<45) = 0.31$



$$\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{45-\mu}{\sigma}\right) = 0.31$$

$$\Rightarrow P\left(z<\frac{45-\mu}{\sigma}\right) = 0.31$$

$$\Rightarrow 0.5 - P\left(\frac{45-\mu}{\sigma}<z<0\right) = 0.31$$

$$\Rightarrow P\left(\frac{45-\mu}{\sigma}<z<0\right) = 0.19$$

$\Rightarrow$ here $\frac{45-\mu}{\sigma}<0$, & area $= 0.19$ corresponds to z score 0.50

ie $$\boxed{\frac{45-\mu}{\sigma} = -0.5} \text{ --- } (2)$$

By equation (1) & (2)

$$64-\mu = (1.41)\sigma \text{ --- } (3)$$

$$45-\mu = (-0.5)\sigma \text{ --- } (4)$$

$$\boxed{\begin{array}{l} \text{mean} = 49.97 \simeq 50 \\ \text{varrence} = 9.94 \simeq 10 \end{array}}$$

ids

Q-18 A coin is tossed 12 times. find the probability, both exactly and by fitting a normal distribution of getting (a) 4 heads (b) at most 4 heads.

(Given that $\phi(1.44) = 0.4251$, $\phi(0.866) = \cancel{0.051}$ 0.3078

Sol

(1) exact soln ⇒

using Binomial distribution ⇒ $\left[P(X=x) = {}^nC_x\, p^x q^{n-x} \right]$

$n = 12$

$p = 0.5$

$q = 0.5$

(a) 4 heads ⇒

$$P(X=4) = {}^{12}C_4 (0.5)^4 (0.5)^{12-4}$$

$$= \boxed{0.12085}$$

(b) at most 4 heads ⇒

$$P(X \leq 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$$

$$= {}^{12}C_0 (0.5)^0 (0.5)^{12} + {}^{12}C_1 (0.5)^1 (0.5)^{11}$$

$$+ {}^{12}C_2 (0.5)^2 (0.5)^{10} + {}^{12}C_3 (0.5)^3 (0.5)^9$$

$$+ {}^{12}C_4 (0.5)^4 (0.5)^{10}$$

$$= \boxed{0.19384}$$

(2) normal distribution ⇒

$\mu = np = 12 \times 0.5 = 6$

$\sigma^2 = npq = 12 \times 0.5 \times 0.5 = 3$

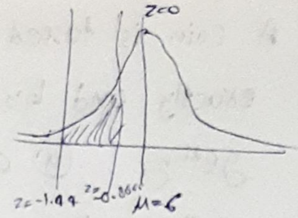
$\sigma = 1.732$

(a) 4 heads ⇒

$P(X=4) = ?$

let $P(3.5 < X < 4.5) = P\left(\frac{3.5-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{4.5-\mu}{\sigma}\right)$

$\Rightarrow P\left(\frac{3.5-6}{1.732} < Z < \frac{4.5-6}{1.732}\right)$



$\Rightarrow P(-1.44 < Z < -0.8666)$

$\Rightarrow P(-1.44 < Z < 0) - P(-0.8666 < Z < 0)$

$\Rightarrow 0.4251 - 0.3078$

$\Rightarrow 0.1173 \simeq \boxed{0.12}$ Ans

(b) at most 4 heads.

~~P(X ≥ 4)~~

$P(X \le 4) \Rightarrow$ let $P(X \le 4.5)$

$\Rightarrow P\left(\frac{x-\mu}{\sigma} \le \frac{4.5-\mu}{\sigma}\right)$

$\Rightarrow P\left(Z \le \frac{4.5-6}{1.732}\right)$

$\Rightarrow P(Z \le -0.866)$

$\Rightarrow 0.5 - P(-0.866 < Z < 0)$

$\Rightarrow 0.5 - 0.3078$

$\Rightarrow \boxed{0.1922}$ Ans

Q-19 The number of personal computers (PCs) sold daily at Computer World is Uniformly distributed with a minimum of 2000 PCs and a maximum of 5000 PCs

(i) Find the probability that the daily sales will fall between 2500 and 3000 PCs.

(ii) What is the probability that Compu. World will sell at least 4000 PCs and exactly 2500 PCs.?

Sol

$$PDF = f(x) = \begin{cases} \frac{1}{5000-2000} = \frac{1}{3000} & ; \quad 2000 < x < 5000 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

(i) $P(2500 < x < 3000) = \int_{2500}^{3000} \frac{1}{3000} \cdot dx$

$= \frac{1}{3000} [3000 - 2500]$

$= \frac{500}{3000}$

$= \boxed{0.1667}$ Ans

(ii) (a) Sell at least 4000 Pcs ⇒

$P(x \geq 4000) = \int_{4000}^{5000} \frac{1}{3000} \cdot dx$

$= \frac{1}{3000} [5000 - 4000]$

$= \frac{1000}{3000}$

$= \boxed{0.3333}$ Ans

(b) exactly 2500 Pc ⇒

$P(x = 2500) = 0$ { in Continuous Random variable, Probability at discrete points is zero }

Q-20 The length of a time a person speaks over phone follows exponential distributions with mean 6. What is the probability that the person will talk for

(i) more than 8 minutes

(ii) between 4 and 8 minutes.

Sol

$\mu = 6 = \frac{1}{\lambda}$

$\Rightarrow \boxed{\lambda = \frac{1}{6}}$

→ exponential distribution PDF ⇒

$f(x) = \begin{cases} e^{-\lambda x} \lambda & ; x \geq 0 \\ 0 & ; \text{elsewhere} \end{cases}$

(i) $P(x > 8) = \int_{8}^{\infty} e^{-\lambda x} \lambda \cdot dx$

$= \int_{8}^{\infty} e^{-(1/6)x} \left(\frac{1}{6}\right) dx = \frac{1}{6} \left[\frac{e^{-x/6}}{-1/6} \right]_{8}^{\infty} = -\left[e^{-\infty} - e^{-8/6}\right]$

$= e^{-8/6}$

$= \boxed{0.2636}$ Ans

(ii) $P(4 < x < 8) = \int_4^8 e^{-x/6}\left(\frac{1}{6}\right) dx$

$= \frac{1}{6}\left[\frac{e^{-x/6}}{-1/6}\right]_4^8$

$= -\left[e^{-8/6} - e^{-4/6}\right]$

$= \boxed{0.2498}$ Ans